

# Leo Zhang

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## PROFESSIONAL SUMMARY

Master's in Computer and Information Sciences from AUT, **First Class Honours**. Senior Software Engineer — Software engineer with 7+ years building IoT platforms, gateway **firmware** and backend **systems** for industrial environments. Experienced in C/C++ on **Linux**, low-level protocol implementation and real-time sensor data processing across serial, GPIO and network interfaces. Delivered gateway software and device connectivity for fleet-scale IoT deployments spanning 10+ factory sites, handling protocol translation and cloud uplink. Practical problem-solver who bridges **hardware** constraints with software design, focused on reliability and maintainability in complex **systems**.

## CORE COMPETENCIES

**Systems Programming:** C, **Linux**, **Embedded Linux**, Git

**IoT & Embedded:** MQTT, UART/Serial, GPIO, Protocol Implementation

**Communication Protocols:** TCP/IP, HTTP, WebSocket, RS232/RS485

**Software Engineering:** SOLID Principles, Design Patterns, CI/CD, Docker

**Backend & Integration:** Java, Spring Cloud, MySQL, Redis

## EXPERIENCE

**Auckland University of Technology** — *Auckland, New Zealand*

2024 – 2026

### AI Research Engineer

- Designed and delivered ChatClothes, a digital multimodal AI platform integrating computer vision, diffusion models, and LLM-powered workflows with RAG, vector search, and prompt grounding. Provided technical leadership and owned the **product** end-to-end as cloud-native AI solutions, from system design and development through evaluation, deployment, and user validation.
- Deployed and optimized AI models on edge devices including Raspberry Pi and Android phones, running lightweight LLMs and vision models on resource-constrained ARM **hardware** with offline inference and on-device prediction.

**Tech:** Python, PyTorch, Ollama, Docker, Linux, AWS, Azure, Git, Raspberry Pi, Android

**Chunxiao Technology Co., Ltd.** — *China*

2013 – 2024

*Technology company (est. 2013, ~40 employees) delivering software, **hardware**, and system integration across streaming media, enterprise platforms, IoT, and industrial automation.*

**Additional Experience** — *Auckland, New Zealand*

### Customer Service & Operations (Part-time)

- Gilmours & Red Cross Shop: customer service, inventory management, and team collaboration

## EDUCATION

**Master of Computer and Information Sciences** *Auckland University of Technology, NZ*

2024 – 2026

Research: Computer Vision, Diffusion Models, Large Language Models, Multimodal AI

**Bachelor of Software Engineering**

*Hebei Univ. of Sci. & Tech., China*

2009 – 2013

Coursework: Data Structures, Algorithms, Java Programming, Database **Systems**, Software **Engineering**

LICENSES & CERTIFICATIONS

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|-------------------------------------------------------------------------------|------|
| IoT Fundamentals — Xi'an Jiaotong University                                  | 2020 |
| Science and Technology Achievement Award (Smart Manufacturing) — Hebei, China | 2020 |
| Software Design Engineer Certificate — China Qualification Authority          | 2019 |

INTERESTS

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Hiking, self-hosted infrastructure, AI tools, and mobile technology trends

REFEREES

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*References available upon request.*